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Project Name:

Soil Moisture Monitoring System using
Window Comparator and RTL Logic

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Submitted Date: 20 May, 2026

Abstract

This project presents the design and implementation of a purely analog Soil Moisture Monitoring System, developed to address the critical need for efficient water management in agriculture and gardening. Unlike conventional digital solutions that rely on microcontrollers or complex integrated circuits, this system utilizes fundamental analog electronics to provide real-time visual feedback on soil conditions. A soil moisture probe, configured within a dynamic voltage divider network, measures the variable resistance of the soil based on its water content. The core decision-making process is executed by an LM358 operational amplifier configured as a window comparator. This comparator evaluates the dynamic input voltage against manually calibrated upper (3.7V) and lower (2.0V) thresholds. Furthermore, a custom Resistor-Transistor Logic (RTL) NOR gate, constructed entirely from discrete BC547 NPN transistors, is implemented to indicate the optimal moisture state. The system successfully and reliably classifies the soil condition into Dry (Red LED), Wet (Blue LED), and Optimum (Green LED) states. By eliminating the need for microcontrollers, this low-cost, energy-efficient, and easily maintainable project effectively bridges theoretical analog laboratory concepts with practical environmental conservation, offering a highly accessible tool for smart agriculture.

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1 Introduction

Proper irrigation is a critical factor in agriculture, gardening, and overall plant maintenance. Both under-watering and over-watering can severely hinder plant growth, cause root rot, and lead to the unnecessary wastage of valuable water resources. In recent times, many modern monitoring systems rely heavily on digital microcontrollers, such as Arduino or microprocessors, to track soil conditions. However, these digital solutions can often be unnecessarily complex, power-consuming, and expensive for fundamental monitoring tasks.

To address this issue, this project introduces a purely analog “Soil Moisture Monitoring System.” The proposed system provides real-time visual feedback on soil conditions without requiring any microcontrollers or digital integrated circuits (like standard logic ICs). It operates by continuously measuring the electrical resistance of the soil using a moisture probe configured within a voltage divider network.

The core decision-making process is executed by an LM358 Dual Operational Amplifier configured as a “Window Comparator.” This setup establishes an adjustable voltage “window” that represents the optimal moisture level. To further process the comparator’s output and indicate the perfect soil condition, the system incorporates Resistor-Transistor Logic (RTL) to construct a custom NOR gate entirely from discrete NPN transistors (BC547). By combining fundamental analog components, this project bridges the gap between theoretical laboratory concepts and practical, real-world applications, offering a reliable, cost-effective, and educational solution for smart moisture monitoring.

2 Project Objectives

- i. To demonstrate the practical implementation of operational amplifiers as window comparators using the LM358 IC and analyze voltage divider networks in sensor interfacing.
- ii. To design and implement a microcontroller-free decision-making circuit using Resistor-Transistor Logic (RTL) with discrete BC547 NPN transistors to build a functional NOR gate.
- iii. To bridge the gap between theoretical electronic concepts learned in the laboratory and real-world engineering challenges by developing a reliable, low-cost soil monitoring tool.
- iv. To promote environmental sustainability and smart agriculture by preventing water wastage and protecting plants from the adverse effects of over-watering and under-watering.

3 Required Components and Resources (Apparatus)

The implementation of this analog soil moisture monitoring system requires a specific set of hardware components and software tools. The resources utilized for designing, simulating, and building the project are listed in Table 1.

Table 1: List of Required Components and Resources

Sl. No.	Component	Specification	Qty	Purpose in the Circuit
i	Operational Amplifier	LM358 (Dual Op-Amp)	1	Core window comparator.
ii	NPN Transistor	BC547	2	Construct custom RTL NOR gate.
iii	Soil Moisture Probe	Analog Resistive Type	1	Detects moisture levels.
iv	LED Indicators	Red, Blue, Green	3	Visual feedback for states.
v	Potentiometers	10k Ω Trimpot	2	Set reference voltages.
vi	Resistors (Base)	47k Ω	2	Base current limiting.
vii	Resistors	220 Ω , 1k Ω	Multiple	Current limiting for LEDs.
viii	Power Supply	5V - 9V DC	1	Power analog circuit.
ix	Breadboard & Wires	Standard	As req.	Prototyping.
x	Software Tool	Proteus Design Suite	1	Circuit simulation.

4 Working Principle and Methodology

The proposed analog soil moisture monitoring system operates based on fundamental electronic principles. The entire methodology can be divided into three distinct functional stages: the sensory stage, the decision-making stage, and the logic indication stage.

4.1 Stage 1: Sensor and Voltage Divider Network

The soil moisture probe functions as a variable resistor (R_s) whose resistance changes inversely with the moisture content of the soil. This probe is connected in series with a fixed resistor to form a basic voltage divider network.

- i. **Dry Soil:** Dry soil acts as a poor conductor, resulting in a high resistance. This causes the output voltage from the divider network (V_{in}) to be relatively high.
- ii. **Wet Soil:** When water is present, the soil becomes highly conductive, dropping the resistance. Consequently, the output voltage (V_{in}) drops to a lower value.

4.2 Stage 2: The Window Comparator (LM358)

The varying input voltage (V_{in}) is fed into a “Window Comparator” circuit built using the LM358 dual operational amplifier. Two potentiometers are used to set two reference

voltages: an upper threshold (V_{high}) and a lower threshold (V_{low}). The op-amps compare V_{in} against these thresholds:

- i. **Dry Condition Indicator:** Op-Amp 1 is configured to compare V_{in} with V_{high} . If the soil is too dry ($V_{in} > V_{high}$), the op-amp outputs a HIGH signal, turning ON the **Red LED**.
- ii. **Wet Condition Indicator:** Op-Amp 2 compares V_{in} with V_{low} . If the soil is excessively wet ($V_{in} < V_{low}$), this op-amp outputs a HIGH signal, turning ON the **Blue LED**.

4.3 Stage 3: RTL NOR Gate (Optimum Condition Indicator)

When the soil moisture is perfectly balanced, the input voltage falls within the defined window ($V_{low} \leq V_{in} \leq V_{high}$). During this state, both Op-Amp 1 and Op-Amp 2 output a LOW (0) signal. To visually indicate this optimum state, a custom Resistor-Transistor Logic (RTL) NOR gate is implemented using two discrete BC547 NPN transistors connected in parallel.

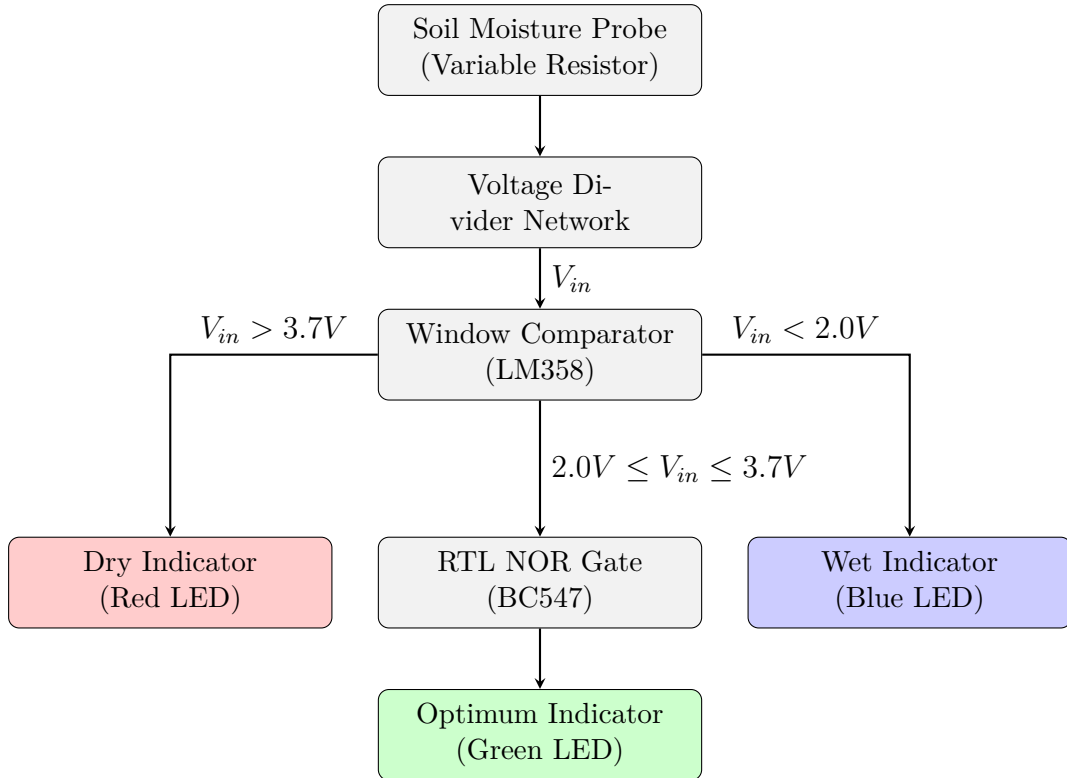


Figure 1: Operational Flowchart of the Soil Moisture Monitoring System

5 Circuit Diagram & Threshold Calibration and Logic Analysis

5.1 Circuit Diagram

The complete schematic design of the Soil Moisture Monitoring System was developed and verified using the Proteus Design Suite. The circuit integrates the LM358 operational

amplifier, two BC547 NPN transistors forming the RTL NOR gate, and the voltage divider networks for both the sensor and threshold references.

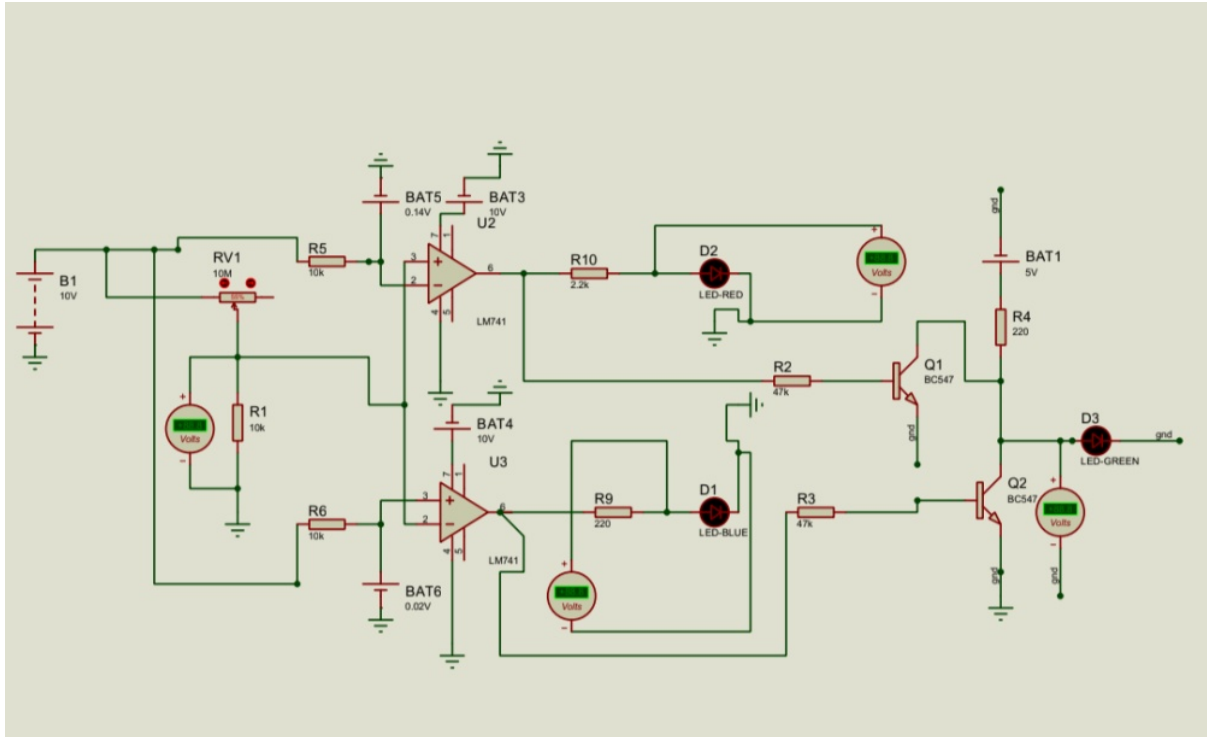


Figure 2: Circuit Diagram of the Soil Moisture Monitoring System

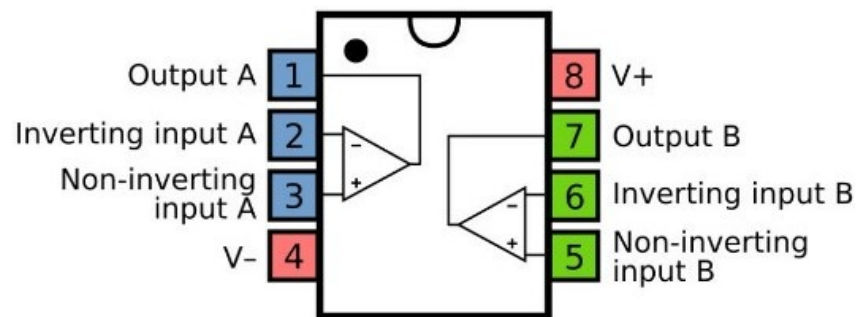


Figure 3: Pin diagram of LM-358-IC

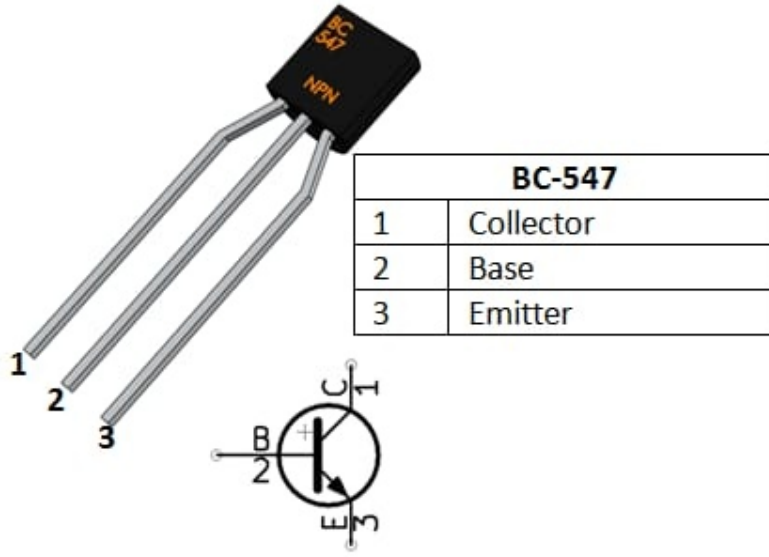


Figure 4: Pin diagram of BC-547

5.2 Threshold Calibration and Logic Analysis

Since the soil moisture probe acts as a variable resistor (R_s) whose value constantly fluctuates based on the physical water content in the soil, providing a static mathematical calculation for the input voltage (V_{in}) is not practical. Instead, the system relies on a dynamic voltage divider network formed by the moisture probe and a fixed $10k\Omega$ base resistor (R_1).

To calibrate the window comparator for accurate decision-making, two reference voltages were manually set using potentiometers:

- i. **Upper Threshold (V_{high}):** Calibrated to **3.7V**. This voltage limit represents the maximum allowable dryness of the soil.
- ii. **Lower Threshold (V_{low}):** Calibrated to **2.0V**. This voltage limit represents the maximum allowable wetness (over-watered state).

The LM358 operational amplifiers continuously evaluate the dynamic input voltage (V_{in}) against these fixed thresholds to determine the exact state of the soil. The logic operates as follows:

- i. **Dry State (Red LED ON):** When the soil dries up, it acts as a poor conductor, causing its resistance to increase. This raises the input voltage. If the voltage exceeds the upper limit ($V_{in} > 3.7V$), Op-Amp 1 triggers the Red LED.
- ii. **Wet State (Blue LED ON):** When the soil is over-watered, it becomes highly conductive, causing a sharp drop in resistance. If the voltage falls below the lower limit ($V_{in} < 2.0V$), Op-Amp 2 triggers the Blue LED.
- iii. **Optimum State (Green LED ON):** When the soil moisture is perfectly balanced, the input voltage naturally fluctuates safely inside the set window (**Between 2.0V and 3.7V**). During this state, both comparators output a LOW signal, and the RTL NOR gate successfully activates the Green LED.

6 Simulation and Troubleshooting

6.1 Proteus Simulation Verification

Before proceeding to the physical hardware implementation on a breadboard, the entire circuit was simulated using the Proteus Design Suite. This step was crucial for analyzing the logical flow, observing real-time voltage variations across the divider networks, and identifying potential design flaws in the analog circuitry.

6.2 Troubleshooting and Design Modifications

During the simulation phase, several technical anomalies were identified and subsequently resolved to ensure the system's practical viability. These modifications demonstrate a deep analysis of the circuit's behavior:

- i. **Op-Amp Selection and Power Supply Optimization:** The initial circuit draft utilized the LM741 operational amplifier. However, the LM741 is an older IC that typically requires a dual power supply (e.g., +10V and -10V) to function accurately. Implementing a dual supply with negative voltages is highly impractical for a portable, simple battery-operated soil monitoring system. To resolve this, the IC was replaced with the **LM358** dual op-amp. The LM358 is specifically optimized for single-supply operation (e.g., 5V), completely eliminating the need for complex negative voltage configurations.
- ii. **Correction of Comparator Logic Inversion:** In the early simulation stages, a logical error occurred due to inverted pin connections. The sensor's voltage was fed into the inverting pin (-) for the dry indicator and the non-inverting pin (+) for the wet indicator, which caused the LEDs to trigger oppositely (e.g., the Blue LED activated when the soil was dry). This was resolved by swapping the inputs according to the correct window comparator logic: the sensor output is now connected to the non-inverting (+) pin of Op-Amp 1 (Dry state) and the inverting (-) pin of Op-Amp 2 (Wet state).
- iii. **Resistor Value Standardization:** The simulation revealed inconsistencies in the chosen component values. Initially, a $10\text{M}\Omega$ potentiometer was used to simulate the sensor, which caused excessive voltage drops and made threshold calibration nearly impossible. This was corrected by assuming a standard $10\text{k}\Omega$ to $100\text{k}\Omega$ range to closely mimic an actual physical soil probe. Furthermore, the current-limiting resistors for the LEDs were initially unbalanced ($2.2\text{k}\Omega$ for the Red LED and 220Ω for the Blue LED), resulting in uneven brightness. To ensure uniform visibility across all indicators, the resistors for the Red, Blue, and Green LEDs were standardized to 220Ω .

7 Environmental and Societal Impact

The development of this purely analog Soil Moisture Monitoring System carries significant implications for both environmental conservation and societal advancement, particularly in agricultural sectors.

- i. **Water Conservation:** Water scarcity is a growing global concern. In traditional farming and gardening, water is often applied based on estimates, leading to massive water wastage through over-irrigation. This system provides an exact, real-time visual cue of the soil's moisture content, ensuring that water is only used when the soil is genuinely dry (Red LED state). This drastically reduces the unnecessary consumption of freshwater resources.
- ii. **Agricultural Efficiency and Plant Health:** Over-watering depletes the oxygen supply in the soil, leading to root rot and the destruction of crops, while under-watering stunts plant growth. By clearly indicating the "Optimum" state (Green LED), this tool helps farmers and gardeners maintain the perfect balance, thereby increasing agricultural yield, improving food security, and reducing financial losses.
- iii. **Low E-Waste and Accessibility:** Unlike complex digital systems that require microcontrollers, extensive coding, and frequent maintenance, this analog system is highly durable, extremely low-cost, and easy to repair. This makes the technology highly accessible to local farmers with limited technical knowledge and financial resources. Furthermore, relying on discrete analog components significantly reduces the electronic waste (e-waste) footprint compared to discarding complex digital circuit boards.

8 Conclusion

The objective of designing and analyzing a functional, purely analog Soil Moisture Monitoring System was successfully achieved. By utilizing an LM358 operational amplifier configured as a window comparator and employing basic NPN transistors (BC547) to create an RTL NOR gate, the project successfully demonstrated how complex decision-making processes can be executed without the use of microcontrollers. The system reliably categorized the soil state into Dry, Wet, and Optimum conditions, providing immediate visual feedback through respective LEDs. Troubleshooting the initial design—such as replacing the LM741 with the LM358 for single-supply operation and correcting the comparator logic—highlighted the importance of in-depth circuit analysis and simulation. Ultimately, this low-cost, energy-efficient project serves as a highly practical tool for smart agriculture and water conservation. Future scopes for this project could include replacing the 'Dry' indicator LED with an automated water pump or buzzer for automated irrigation control.

9 References

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